```
PS C:\Users\B R MANAVI\Downloads\L5_39_vehicle_service-main> Get-ChildItem vehicle_service_original.py


    Directory: C:\Users\B R MANAVI\Downloads\L5_39_vehicle_service-main


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        13-10-2026     05:01           2337 vehicle_service_original.py
```

pylint before

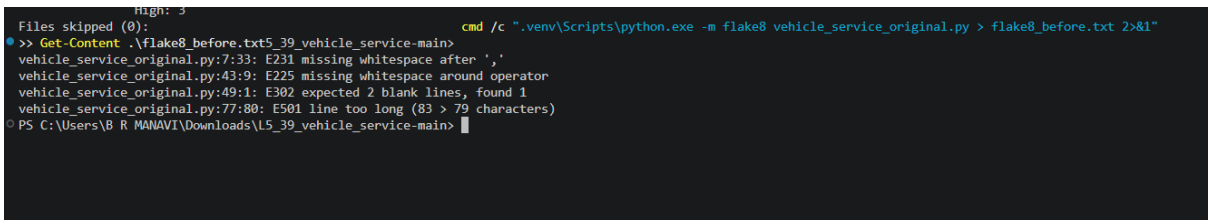
```
PS C:\Users\B R MANAVI\Downloads\L5_39_vehicle_service-main> Get-Content .\pylint_before.txt
************* Module vehicle_service_original
vehicle_service_original.py:1:0: C0114: Missing module docstring (missing-module-docstring)
vehicle_service_original.py:12:0: C0116: Missing function or method docstring (missing-function-docstring)
vehicle_service_original.py:12:0: C0103: Function name "addService" doesn't conform to snake_case naming style (invalid-name)
vehicle_service_original.py:12:15: C0103: Argument name "vehicleNo" doesn't conform to snake_case naming style (invalid-name)
vehicle_service_original.py:12:26: C0103: Argument name "serviceType" doesn't conform to snake_case naming style (invalid-name)
vehicle_service_original.py:12:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
vehicle_service_original.py:20:0: C0116: Missing function or method docstring (missing-function-docstring)
vehicle_service_original.py:20:0: C0103: Function name "cancelService" doesn't conform to snake_case naming style (invalid-name)
vehicle_service_original.py:20:18: C0103: Argument name "vehicleNo" doesn't conform to snake_case naming style (invalid-name)
vehicle_service_original.py:20:29: C0103: Argument name "serviceType" doesn't conform to snake_case naming style (invalid-name)
vehicle_service_original.py:23:11: W0718: Catching too general exception Exception (broad-exception-caught)
vehicle_service_original.py:31:0: C0116: Missing function or method docstring (missing-function-docstring)
vehicle_service_original.py:31:0: C0103: Function name "getTotalCost" doesn't conform to snake_case naming style (invalid-name)
vehicle_service_original.py:31:17: C0103: Argument name "vehicleNo" doesn't conform to snake_case naming style (invalid-name)
vehicle_service_original.py:32:4: C0103: Variable name "totalCost" doesn't conform to snake_case naming style (invalid-name)
vehicle_service_original.py:34:8: C0103: Variable name "totalCost" doesn't conform to snake_case naming style (invalid-name)
vehicle_service_original.py:38:0: C0116: Missing function or method docstring (missing-function-docstring)
vehicle_service_original.py:38:0: C0103: Function name "jobCardId" doesn't conform to snake_case naming style (invalid-name)
vehicle_service_original.py:38:14: C0103: Argument name "vehicleNo" doesn't conform to snake_case naming style (invalid-name)
vehicle_service_original.py:42:0: C0116: Missing function or method docstring (missing-function-docstring)
vehicle_service_original.py:42:0: C0103: Function name "busiestVehicle" doesn't conform to snake_case naming style (invalid-name)
vehicle_service_original.py:49:0: C0116: Missing function or method docstring (missing-function-docstring)
vehicle_service_original.py:49:0: C0103: Function name "saveRecords" doesn't conform to snake_case naming style (invalid-name)
vehicle_service_original.py:49:16: C0103: Argument name "fileName" doesn't conform to snake_case naming style (invalid-name)
vehicle_service_original.py:50:9: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
vehicle_service_original.py:54:0: C0116: Missing function or method docstring (missing-function-docstring)
vehicle_service_original.py:54:0: C0103: Function name "loadRecords" doesn't conform to snake_case naming style (invalid-name)
vehicle_service_original.py:54:16: C0103: Argument name "fileName" doesn't conform to snake_case naming style (invalid-name)
vehicle_service_original.py:59:0: C0116: Missing function or method docstring (missing-function-docstring)
vehicle_service_original.py:59:0: C0103: Function name "backupRecords" doesn't conform to snake_case naming style (invalid-name)
vehicle_service_original.py:64:0: C0116: Missing function or method docstring (missing-function-docstring)

---------------------------------
Your code has been rated at 4.66/10
```

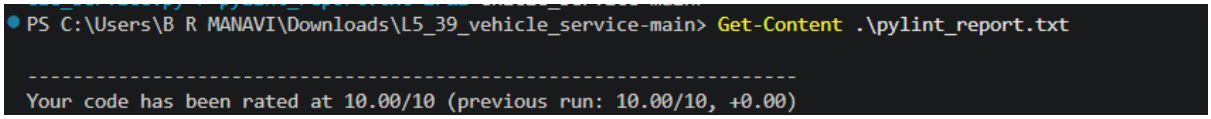
bandit before



flake8 before



Pylint after



Bandit after

```
PS C:\Users\B R MANAVI\Downloads\L5_39_vehicle_service-main> cmd /c ".venv\Scripts\python.exe -m bandit -r vehicle_service.py > bandit_report.txt 2>&1"
PS C:\Users\B R MANAVI\Downloads\L5_39_vehicle_service-main> Get-Content .\bandit_report.txt
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.14.0
Run started:2026-10-14 09:16:04.438311+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 110
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
PS C:\Users\B R MANAVI\Downloads\L5_39_vehicle_service-main>
```

flake8 after

```
PS C:\Users\B R MANAVI\Downloads\L5_39_vehicle_service-main> cmd /c ".venv\Scripts\python.exe -m flake8 vehicle_service.py > flake8_report.txt 2>&1"
PS C:\Users\B R MANAVI\Downloads\L5_39_vehicle_service-main> Get-Content .\flake8_report.txt
PS C:\Users\B R MANAVI\Downloads\L5_39_vehicle_service-main>
```

results

```
PS C:\Users\B R MANAVI\Downloads\L5_39_vehicle_service-main> .\.venv\Scripts\python.exe -m py_compile vehicle_service.py test_vehicle_service.py
PS C:\Users\B R MANAVI\Downloads\L5_39_vehicle_service-main> .\.venv\Scripts\python.exe -m unittest -v
test_add_service_cost_and_total (test_vehicle_service.VehicleServiceTests.test_add_service_cost_and_total) ... ok
test_cancel_existing_service (test_vehicle_service.VehicleServiceTests.test_cancel_existing_service) ... ok
test_cancel_missing_record_returns_false (test_vehicle_service.VehicleServiceTests.test_cancel_missing_record_returns_false) ... No service records found for vehicle KA09ZZ0001
ok
test_default_history_is_not_shared (test_vehicle_service.VehicleServiceTests.test_default_history_is_not_shared) ... ok
test_empty_vehicle_number_is_rejected (test_vehicle_service.VehicleServiceTests.test_empty_vehicle_number_is_rejected) ... ok
test_job_card_id_is_stable_and_short (test_vehicle_service.VehicleServiceTests.test_job_card_id_is_stable_and_short) ... ok
test_save_and_load_records (test_vehicle_service.VehicleServiceTests.test_save_and_load_records) ... ok
test_unknown_service_type_is_rejected (test_vehicle_service.VehicleServiceTests.test_unknown_service_type_is_rejected) ... ok

----------------------------------------------------------------------
Ran 8 tests in 0.020s

OK
PS C:\Users\B R MANAVI\Downloads\L5_39_vehicle_service-main>
```

final program

```
PS C:\Users\B R MANAVI\Downloads\L5_39_vehicle_service-main> .\.venv\Scripts\python.exe vehicle_service.py
WARNING: No service records found for vehicle KA09ZZ0001
Total for KA01AB1234: 2300
Total for KA05XY9876: 2500
Job card for KA01AB1234: bd2cd41f
Vehicles on file: 3 | busiest vehicle: KA01AB1234
PS C:\Users\B R MANAVI\Downloads\L5_39_vehicle_service-main>
```